

CLASS 6 — WEEK 2

Metabolism, Part 2

Building, storing, regulating, feeding

Dr. Giorgio Lagna • Wednesday, April 22, 2026

BIOL 40C • Human A&P III • Foothill College

Warm-up: What did we cover on Monday?

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ACTIVITY

Neighbor recall (30 seconds, no notes): Name the three stages of glucose catabolism and where each happens (which compartment of the cell).

Expected answer:

1. **Glycolysis** → cytosol (anaerobic; glucose → 2 pyruvate)
2. **Citric acid cycle** → mitochondrial matrix (aerobic; CO₂ out, NADH/FADH₂ produced)
3. **Electron transport chain + ATP synthase** → inner mitochondrial membrane (H⁺ gradient → ATP)

Flip the frame: Monday we BROKE things down. Today we BUILD them up.

Today's roadmap: four blocks

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CONCEPT

23.4

Anabolic pathways

How do cells BUILD glycogen, fat, and protein?

23.5

Metabolic states & feeding

Insulin/glucagon, absorptive vs. postabsorptive, satiety

23.6

Metabolic rate & thermoregulation

BMR, thyroid, heat balance, shivering

23.7

Nutrition & body mass

Macronutrients, vitamins, lipoproteins, Juan's celiac callback

Catabolism vs. Anabolism: the same logic, flipped

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CONCEPT

CATABOLISM (Monday)

Breaking DOWN

Large molecules → small molecules

RELEASES energy → captured as ATP

Exergonic ($\Delta G < 0$)

Examples:

- Glucose → $\text{CO}_2 + \text{H}_2\text{O} + \text{ATP}$
- Triglyceride → fatty acids → ATP
- Glycogen → glucose (glycogenolysis)

Dominates when: *FASTED / low fuel*

ANABOLISM (Today)

Building UP

Small molecules → large molecules

REQUIRES energy → spends ATP + NADPH

Endergonic ($\Delta G > 0$)

Examples:

- Glucose → glycogen (glycogenesis)
- Acetyl-CoA → fatty acids (lipogenesis)
- Lactate/glycerol → glucose (gluconeogenesis)

Dominates when: *FED / post-meal*

MODULE 23.4

Anabolic Pathways

Not just catabolism in reverse

Why anabolism ≠ running catabolism backwards

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CONCEPT

Core principle: Catabolism releases energy. Anabolism needs energy. *You can't just run the same reactions backwards.* Anabolic pathways use different enzymes and spend ATP + NADPH.

Glycogenesis

Where: Liver + muscle

Glucose → glycogen
(when blood glucose is HIGH)

Stimulated by: *Insulin*

Gluconeogenesis

Where: Liver (+ some kidney)

Lactate/glycerol/AAs → glucose
(when blood glucose is LOW)

Stimulated by: *Glucagon*

Lipogenesis

Where: Liver + adipose

Excess acetyl-CoA → fatty acids → triglycerides

Stimulated by: *Insulin*

Gluconeogenesis: the liver makes glucose from scratch

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CONCEPT

Why does this matter?

Your brain burns ~120 g glucose/day.
During a fast, your glycogen stores last ~16 hours.

After that?

The liver makes NEW glucose from non-carb precursors:

- Lactate (from muscle)
- Glycerol (from triglycerides)
- Glucogenic amino acids (from protein breakdown)

Metformin (remember Steve!) partly works by suppressing hepatic gluconeogenesis.

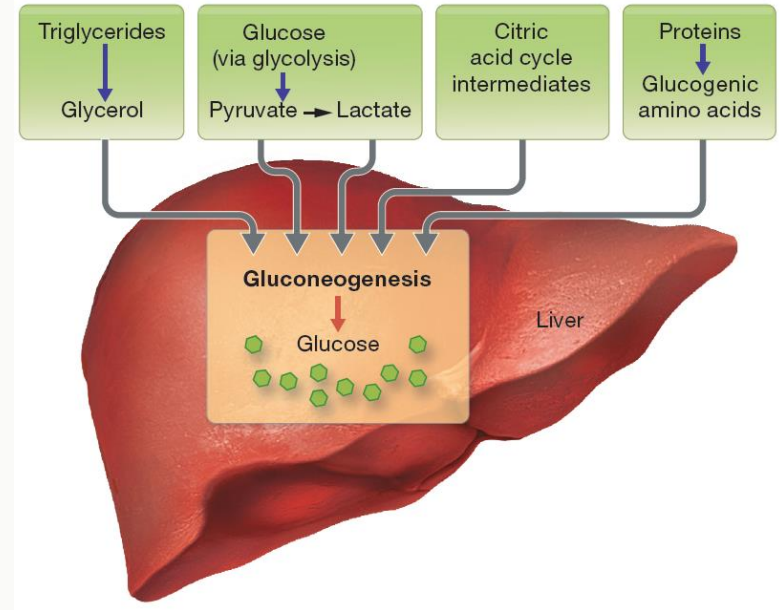


Fig. 23.17 — Precursors feed the liver's gluconeogenesis

Quick poll: where does excess pizza go?

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CHECK

A sedentary student eats 4,000 kcal of pizza. The carbs and protein they don't immediately burn will end up stored as...

A Glycogen only

B Fat only

C Both glycogen and fat

D Amino acid pools

A sedentary student eats 4,000 kcal of pizza. The carbs and protein they don't immediately burn will end up stored as...

Glycogen only

0%

Fat only

0%

Both glycogen and fat

0%

Amino acid pools

0%

Answer: C — Both glycogen and fat

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CHECK



Glycogen fills first (cap ~500 g total). Excess beyond that → fat via lipogenesis.

The logic:

1. **Glycogen storage caps out.** ~100 g in liver, ~400 g in muscle. Once full, you can't add more.
2. **Fat storage essentially has no cap.** Adipocytes expand indefinitely. This is why fat stores grow with surplus.
3. **Excess protein also becomes fat.** The body cannot stockpile amino acids. Carbon skeletons → acetyl-CoA → lipogenesis.

Clinical relevance: *This is the molecular reason calorie surplus accumulates easily and calorie restriction is hard — your biology is optimized for the last famine.*

Brown Bag case: Mark's Low-T booster

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CASE STUDY

Mark, 42, low-carb (keto) for 3 months. He's now taking a Low-T booster that claims to 'help build muscle and cholesterol production.' *Does the claim make sense?*

Think-Pair-Share — work through the biochemistry:

1. Steroid hormones (testosterone) are synthesized from cholesterol.
2. Cholesterol is synthesized from acetyl-CoA (via HMG-CoA reductase).
3. On a keto diet, acetyl-CoA IS abundant — but where is it going?

RCS takeaway: *This is exactly the reasoning the RCS Quiz rewards — mechanism + context.*



TESTOSTERONE BOOSTERS, KETO DIETS, AND CHOLESTEROL:

DOES THE CLAIM MAKE SENSE?



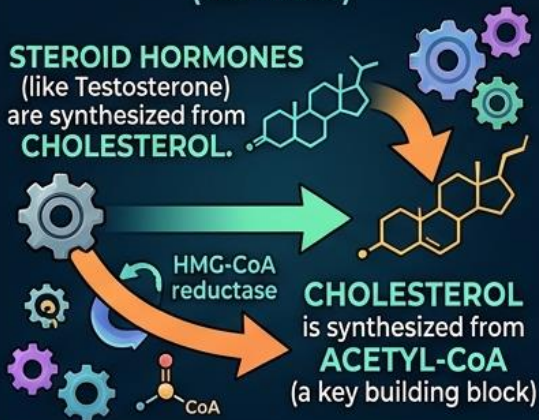
MARK, 42,

!?

MARK, 42, LOW-CARB (KETO) DIET
for 3 months. Takes this Low-T booster.
Does the claim hold up?

HOW HORMONES ARE MADE (The Basics)

STEROID HORMONES
(like Testosterone)
are synthesized from
CHOLESTEROL.



MODULE 23.5

Metabolic States & Feeding

The insulin/glucagon switch flips every few hours

A&P review: negative feedback (from 40A)

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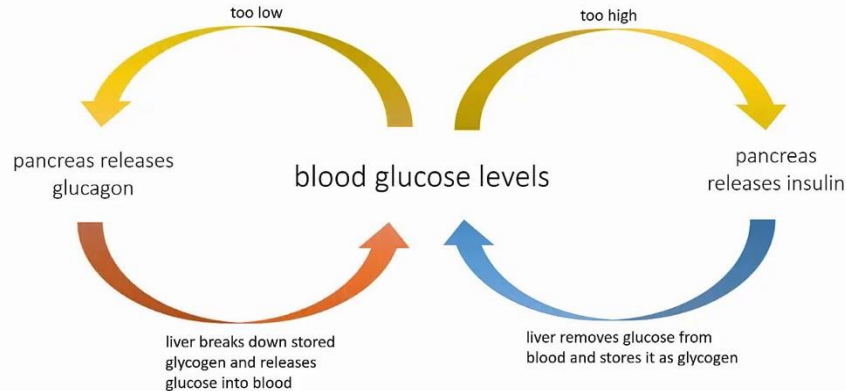
A&P REVIEW

You saw this last quarter. Every hormonal system you'll meet this quarter uses it. Today's example: blood glucose homeostasis.

BLOOD GLUCOSE FALLS

(e.g., between meals, fasting)

1. ↓ Glucose → pancreatic α -cells release **GLUCAGON**
2. → glycogenolysis in liver
3. → gluconeogenesis + lipolysis
4. → *blood glucose rises back to set point.*



BLOOD GLUCOSE RISES

Set point ≈ 90 mg/dL.

1. ↑ Glucose → pancreatic β -cells release **INSULIN**
2. → glucose uptake by tissues
3. → glycogenesis + lipogenesis
4. → *blood glucose falls back to set point.*

ABSORPTIVE state: 0–4 hours post-meal

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CONCEPT

Insulin dominates

The meal floods your blood with glucose, amino acids, and fats.

Insulin tells every tissue:

- Liver: store glucose → glycogen
- Muscle: import glucose + store glycogen
- Adipose: import glucose + store triglyceride

Cells run on **DIETARY** glucose. *Storage mode is ON.*

Net result: *anabolism dominates.*

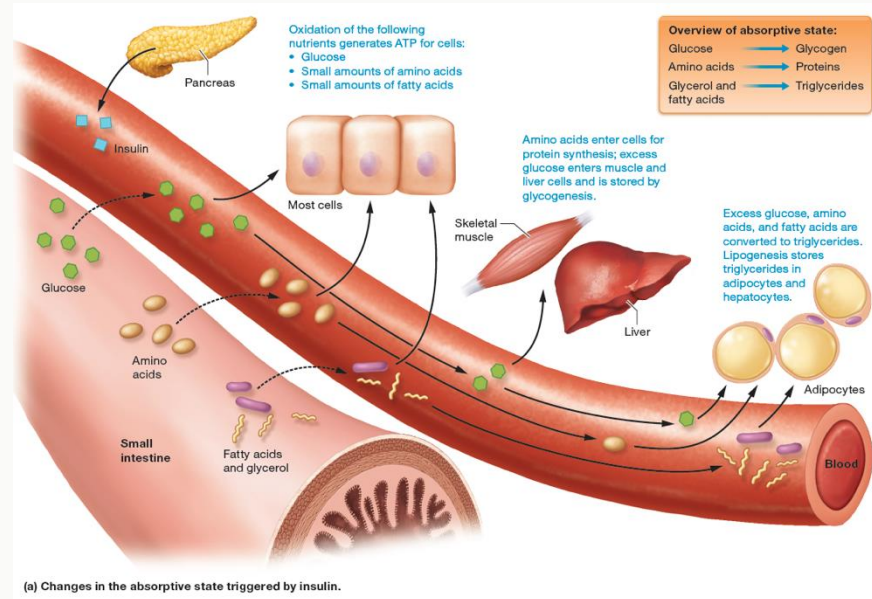


Fig. 23.18a — Absorptive state triggered by insulin

POSTABSORPTIVE state: 4+ hours post-meal

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CONCEPT

Glucagon dominates

Blood glucose has dropped. You're between meals.

Glucagon tells the liver:

- Break down glycogen (glycogenolysis)
- Make NEW glucose (gluconeogenesis)
- Adipose: release fatty acids
- Muscle: release amino acids

Brain + RBCs run on stored glucose. Other tissues shift to fat.

Net result: *catabolism dominates.*

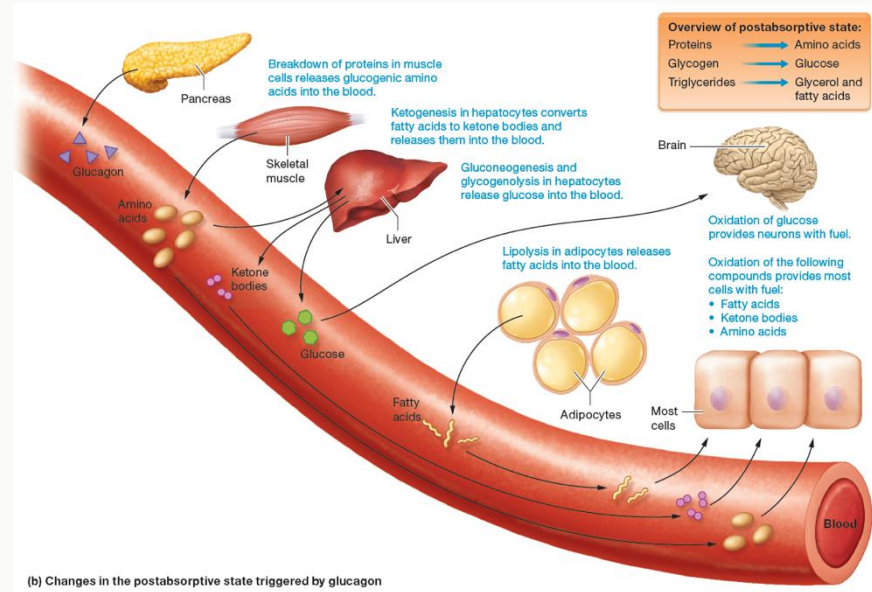


Fig. 23.18b — Postabsorptive state triggered by glucagon

The metabolic states timeline

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CONCEPT

0 – 4 h

Absorptive

Fed state. Insulin dominant. Storing glucose, amino acids, and fats.

4 – 12 h

Postabsorptive

Between meals. Glucagon dominant. Glycogen breakdown feeds blood glucose.

12 – 24 h

Early fasting

Glycogen depleted. Gluconeogenesis ramps up. Fat mobilization accelerates.

>2–3 days

Starvation

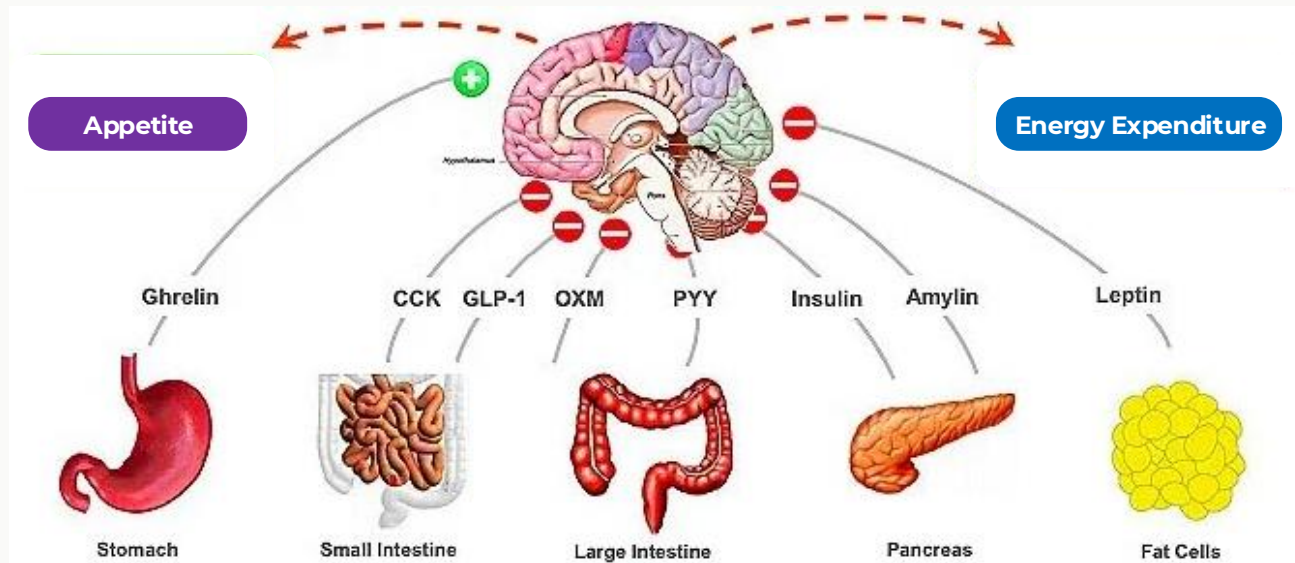
Ketogenesis peaks. Brain adapts to ketones. Protein breakdown slows (adaptive).

KC Q12 tripped students up: *the switch happens around 4 HOURS post-meal, not right after.*

Satiety signals: hormones that drive feeding

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CONCEPT



Ghrelin

Stomach

HUNGER signal. Is a PROTEIN
Peaks BEFORE meals.
→ hypothalamus: EAT!

Leptin

Fat cells

LONG-TERM satiety. Is a PROTEIN
∝ fat mass.
→ reduce intake.

CCK / PYY

Small intestine

SHORT-TERM satiety. Are PROTEINS
Released during meal.
"I'm full" signal.

GLP-1

Small intestine

Satiety + insulin release. Is a PROTEIN
Slows gastric emptying.
Drug target: Wegovy/Ozempic.

Therapeutic connection: *Leptin is HIGH in obesity but the hypothalamus goes resistant (exogenous leptin ≠ weight-loss drug).*

GLP-1 agonists (semaglutide → Wegovy/Ozempic) bypass that by boosting satiety at the hypothalamus and slowing gastric emptying.

Brown Bag case: Travis's supplement shopping

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CASE STUDY

Travis, 21, type 1 diabetic. He wants to lose weight and is considering "leptin" and "ghrelin-blocker" supplements from a website. *What do you tell him?*

Two-layer reasoning (this is RCS-caliber thinking):

Layer 1 — Pharmacology/Digestion: Leptin and ghrelin are peptide hormones. Peptide hormones are digested in the stomach and small intestine (Week 1 content!). *Oral supplements wouldn't survive absorption intact.*

Layer 2 — Endocrinology: Most obesity is associated with leptin RESISTANCE, not deficiency. Travis's endogenous leptin is likely already high. Adding more won't help. *This is why exogenous leptin isn't a weight-loss drug.*

Closing: *Two layers of biology say the product won't work. This kind of layered reasoning is what the RCS Quiz rewards.*

MODULE 23.6

Metabolic Rate & Thermoregulation

Why hypothyroid patients feel cold

Basal Metabolic Rate (BMR)

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CONCEPT

BMR = the energy cost of being alive at rest, post-absorptive, thermoneutral, awake. *Typical adult: ~1,200–1,800 kcal/day. Mostly heat.*

Factors that CHANGE BMR

- **Thyroid hormone** (biggest single regulator)
- **Lean body mass** (muscle burns ~3× fat at rest)
- **Age** (↓ 1–2% per decade after 20)
- Sex, pregnancy, fever, nutrition status.

Total Daily Energy (TDEE)

$$\text{TDEE} = \text{BMR} + \text{TEF} + \text{activity}$$

- BMR = ~60–70% of daily calories
- Thermic effect of food (TEF) ≈ 10%
- Physical activity = variable (0–40%+)

Quick check: *Patient A has more muscle, same weight as B. Whose BMR is higher? (A — muscle is metabolically expensive.)*

Heat balance: four ways your body loses heat

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CONCEPT

Core: held at $\sim 37^{\circ}\text{C}$

Heat production (mostly metabolic) must balance heat loss.

Four loss mechanisms at rest:

- **Radiation** $\approx 60\%$
- **Evaporation** $\approx 22\%$ (more with sweat)
- **Convection** $\approx 15\%$
- **Conduction** $\approx 3\%$

Fever and exercise shift the balance.

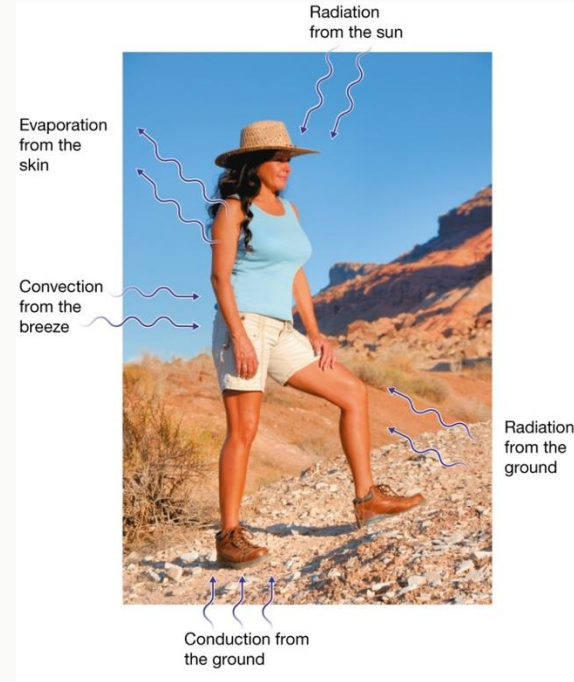
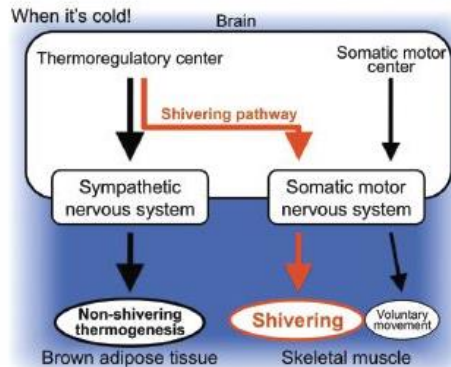
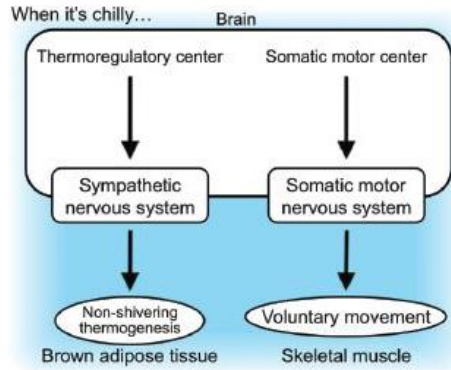


Fig. 23.21 — Four heat-exchange mechanisms

Cold response: shivering vs. non-shivering thermogenesis

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CONCEPT



SHIVERING thermogenesis

Skeletal muscle contracts rapidly and involuntarily.
→ burns glycogen and ATP
→ generates heat as a byproduct of (in)efficient muscle work

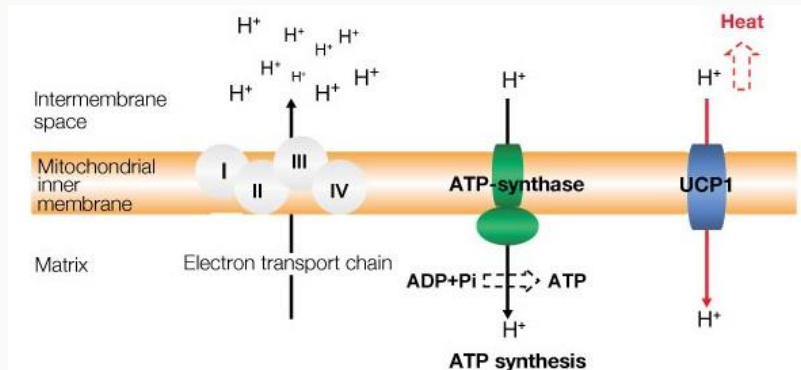
*Dominant in adults.
Short-term cold response.*

KC Q9 target.

NON-SHIVERING thermogenesis

Brown adipose tissue (BAT).
Contains **UCP1 (uncoupling protein 1)** in the inner mitochondrial membrane.
UCP1 lets H^+ flow back across the membrane WITHOUT passing through ATP synthase.
→ Proton gradient collapses → energy released as HEAT, not ATP.

Monday callback: UCP1 is a *natural 2,4-DNP*. One kills. One keeps infants alive.

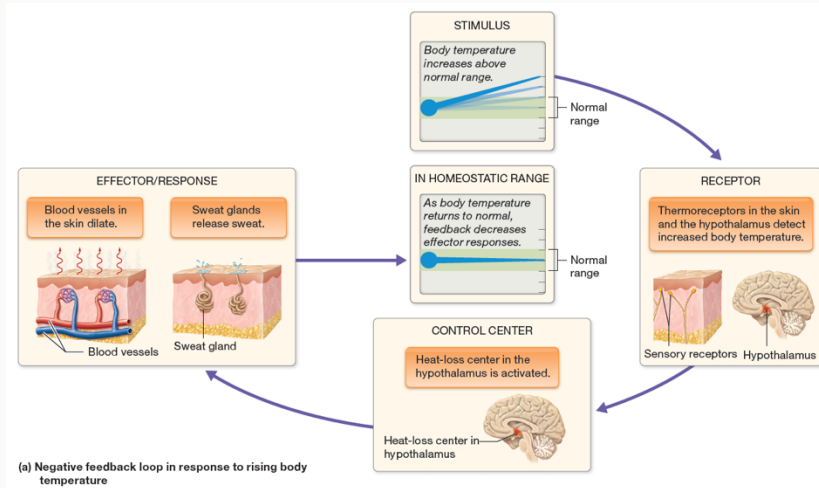


Thermoregulation: hypothalamus drives negative feedback

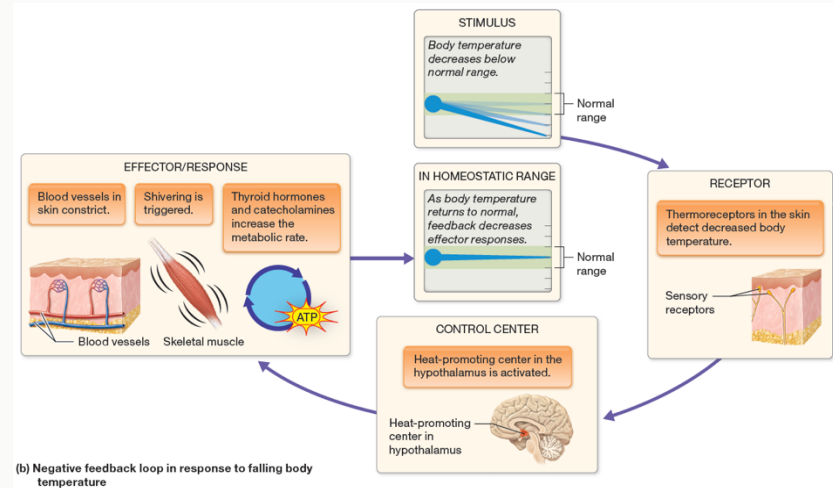
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CONCEPT

Hypothalamic temperature sensors + skin receptors feed into heat-loss OR heat-promoting centers. Two loops, one goal: 37°C.



(a) Response to RISING body temperature



(b) Response to FALLING body temperature

Brown Bag case: Natasha's levothyroxine timeline

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CASE STUDY

Natasha, hypothyroid, started levothyroxine (a synthetic version of the endogenous hormone thyroxine) **3 weeks ago**. She still feels cold, fatigued, and is gaining weight despite eating less. She asks: "Is this medication even working?"

Three questions:

- 1. Which symptoms come from LOW BMR?** Cold intolerance, fatigue, weight gain, slow GI transit. All consistent with ↓ thyroid hormone → ↓ BMR.
- 2. Why is 3 weeks not enough?** Levothyroxine takes 4–6 weeks to reach steady state AND even longer for gene-level effects on mitochondrial biogenesis. She's in the right direction — just not there yet.
- 3. Risk if she doubles her dose on her own?** Iatrogenic thyrotoxicosis — massive ↑ BMR, heat intolerance, tremor, cardiac arrhythmia, bone density loss. *Same risk if Deana stacks her weight-loss supplement on top.*

MODULE 23.7

Nutrition & Body Mass

Macronutrients, lipoproteins, and Juan's callback

The three macronutrients

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CONCEPT

Table 23.1 Summary of Macronutrients

Macronutrient Class	Dietary Sources	Daily Requirement (for Adults)	Functions
Carbohydrates • Monosaccharides and disaccharides (simple sugars) • Polysaccharides (complex carbohydrates)	• Fruits, dairy products, sugar cane, sugar beets, candy, additives to many foods and drinks • Rice, pasta, bread, potatoes, vegetables, glycogen in meat	45–65% of total caloric intake	• Oxidized for energy • Used in production of other substances • Incorporated into multiple structural molecules
Lipids • Saturated • Unsaturated	Meats, eggs, dairy products, plant oils	< 30% of total caloric intake	• Oxidized for fuel • Used to form structural molecules • Glycerol used for gluconeogenesis
Proteins	Meats, legumes, nuts, seeds, dairy products, certain grains, eggs	10–35% of total caloric intake	• Used to form structural molecules • Used to manufacture enzymes, antibodies, and hormones • Oxidized for fuel • Used for gluconeogenesis

Table 23.1 — Carbohydrates, lipids, proteins: sources, RDA, functions

Protein quality: essential amino acids

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CONCEPT

9 essential amino acids

Must come from diet — body can't synthesize them.

Complete proteins have all 9.

- Animal sources (meat, dairy, eggs)
- Soy, quinoa

Incomplete proteins are missing one or more.

- Most plants

Complementary eating solves it (*beans + rice = complete*).

Table 23.2 Essential and Nonessential Amino Acids

Essential Amino Acids	Nonessential Amino Acids
Histidine	Alanine
Isoleucine	Arginine
Leucine	Asparagine
Lysine	Aspartate
Methionine	Cysteine
Phenylalanine	Glutamate
Threonine	Glutamine
Tryptophan	Glycine
Valine	Proline
	Serine
	Tyrosine

Table 23.2 — Essential and non-essential amino acids

Quick poll: the one-food vegan

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CHECK

True or false? *A strict vegan who eats only one plant food group per day (e.g., only rice) would eventually develop protein deficiency, even if total protein intake is adequate.*

True or false: A strict vegan who eats only one plant food group per day (e.g., only rice) would eventually develop protein deficiency, even if total protein intake is adequate.

True

0%

False

0%

Quick poll: the one-food vegan

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CHECK

True or false? *A strict vegan who eats only one plant food group per day (e.g., only rice) would eventually develop protein deficiency, even if total protein intake is adequate.*



TRUE if it's a single incomplete protein.

The body can't stockpile amino acids. If even one essential amino acid is missing, protein synthesis stalls (it's rate-limiting). Excess amino acids get deaminated → carbon skeleton goes to fuel or fat.

Clinical relevance: *kwashiorkor (protein malnutrition) and sarcopenia in the elderly both come down to essential amino acid availability, not just calorie count.*

Lipoproteins: how fats travel in blood

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CONCEPT

Fat can't dissolve in plasma. The liver and intestine package lipids into lipoproteins — protein-coated spheres of fat, cholesterol, and phospholipid.

Chylomicrons

Intestine → lymph → blood · Dietary fat (from meals)

VLDL

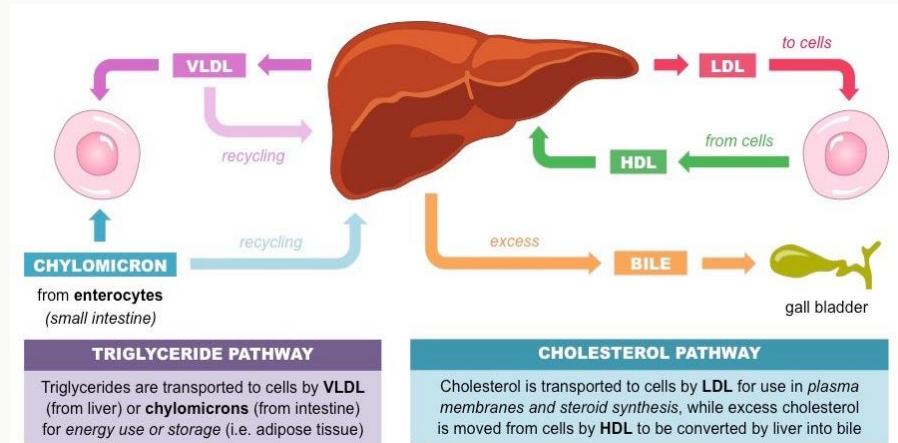
Liver → tissues · Endogenous triglycerides made by liver

LDL ("bad")

Liver → tissues · Cholesterol → tissues.
↑ correlates with atherosclerosis

HDL ("good")

Tissues → liver · Reverse cholesterol transport. Protective



Week 1 callback: Juan's celiac → fat-soluble vitamin deficiency

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CONCEPT

Remember Juan? Celiac disease, damaged intestinal villi. *What happens to his fat absorption — and therefore to his fat-soluble vitamins A, D, E, K?*

Tracing the cascade (this is cross-system A&P reasoning):

1. **Damaged villi** → reduced surface area for absorption
2. **Impaired fat absorption** → reduced chylomicron formation
3. **Vitamins A, D, E, K** ride with fat on chylomicrons — no fat, no delivery.
4. **Clinical consequence:** night blindness (A), osteoporosis (D), neuropathy (E), bleeding (K).

The bigger lesson: one system's pathology (digestion) propagates into another system's nutrition. *That's the A&P lens. That's why we do this.*

Fat-soluble vitamins: A, D, E, K

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CONCEPT

Table 23.3 Summary of Vitamins				
Vitamin	Dietary Sources and RDA	Functions	Symptoms of Deficiency	Symptoms of Excess
Fat-Soluble				
Vitamin A (<i>retinol</i>)	Leafy green vegetables, yellow vegetables; 0.7–0.9 mg	• Required for low-light and color vision • Functions as growth factor for epithelium • Needed for growth and development	• Delayed growth • Night blindness and potential total blindness • Immune dysfunction	• Liver damage • Nausea • Decreased appetite • Osteoporosis • CNS dysfunction • Death from extreme doses
Vitamin D	Synthesized in skin when exposed to certain wavelengths of UV light, supplemented in milk and other foods; 15–20 µg (unless exposure to sunlight is inadequate)	• Required for calcium ion homeostasis and bone growth	• Rickets (in children) • Osteoporosis	• Multiple organ dysfunction due to hypercalcemia
Vitamin E (<i>tocopherol</i>)	Vegetable oils, nuts, seeds, dairy products; 15 mg	• Antioxidant	• Anemia • CNS dysfunction	• Nausea • Fatigue • Vision disturbances
Vitamin K	Leafy green vegetables, produced by intestinal bacteria; 90–120 µg	• Required for synthesis of clotting factors II, VII, IX, and X	• Bleeding disorders	• Liver dysfunction • Excessive clotting in individuals taking the drug warfarin

Table 23.3 (Fat-soluble) — Functions, deficiency, excess

Brown Bag case: Adam's cholesterol panel

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CASE STUDY

Adam, 42. Cholesterol panel: HIGH LDL, LOW HDL. *He's asking: should I start a statin, or try lifestyle changes first?*

What his panel tells us

- **High LDL** = lots of cholesterol being delivered to arteries
- **Low HDL** = less reverse transport out of tissues
- **Net effect:** cholesterol accumulates in vessel walls → plaque → atherosclerosis.

How a statin works

Statin inhibit **HMG-CoA reductase** — the rate-limiting enzyme of cholesterol synthesis.

- Liver makes less cholesterol
- Liver upregulates LDL receptors
- Blood LDL drops

Lifestyle (diet, exercise) also raises HDL — often layered with statins.

Concept check: which state?

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CHECK

It's 11 AM. You ate breakfast at 7 AM. Which hormonal/metabolic state are you in right now?

- A** Absorptive — insulin dominant, storing glycogen
- B** Postabsorptive — glucagon dominant, mobilizing glycogen
- C** Early fasting — gluconeogenesis is ramping up
- D** Starvation — ketogenesis is peaking

It's 11 AM. You ate breakfast at 7 AM. Which hormonal/metabolic state are you in right now?

Absorptive — insulin dominant, storing glycogen

0%

Postabsorptive — glucagon dominant, mobilizing glycogen

0%

Early fasting — gluconeogenesis is ramping up

0%

Starvation — ketogenesis is peaking

0%

Answer: B — Postabsorptive (glucagon dominant)

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CHECK



At 4+ hours post-meal, you're in the postabsorptive state.

Why each option:

- A. Absorptive lasts 0–4 hours. You're past that window.
- B. Postabsorptive = 4–12 h post-meal. Liver glycogen is feeding your blood glucose. ✓
- C. Early fasting starts ~12 h. Not yet.
- D. Starvation = 2–3+ days. You'd know.

Clinical relevance: *this is exactly when a diabetic patient's blood sugar management gets tricky — between-meal glucagon action vs. long-acting insulin.*

CONCEPT

Your body is a hormonally regulated SWITCH.

Insulin + glucagon flip catabolism/anabolism on and off every few hours. *Thyroid hormone sets the dial for how fast the whole system runs.*

If you can...

- ✓ Predict whether insulin or glucagon dominates in a given scenario
- ✓ Draw the absorptive → postabsorptive → fasting → starvation timeline
- ✓ Explain why hypothyroidism causes cold intolerance, fatigue, and weight gain
- ✓ Trace how celiac → impaired fat absorption → fat-soluble vitamin deficiency

...you've nailed Week 2.

CONCEPT

RCS Quiz 2 • Brown Bag Event — six patient vignettes. You've MET all six in class. Use your notes.

Application Activity 2 • Crystal's gallstone case — bridges Week 1 digestive physiology to Week 2 nutrition. After cholecystectomy, which nutrients does she struggle with? Why?

Discussion Board 2 • Share reasoning; respond to two classmates. Credit for both posting and responding.

FOOTSIE / ARIA • Drill anabolic pathways, satiety hormones, and the heat-balance mechanisms.

Next week: *Urinary system. Urea from protein catabolism, metabolic water from the citric acid cycle, and glucose reabsorption all converge in the kidneys. This week feeds forward.*

Muddiest point — before you leave

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ACTIVITY

On your notes, write the ONE thing from this week that's still muddiest for you.

- Could be a pathway you can't quite draw
- Could be a hormone you mixed up
- Could be a Brown Bag patient whose case still feels fuzzy

Ask this question to ARIA or Footsie. If the answer is not yet clear or accurate, email me. I will try to answer it!

Thank you for two weeks of solid work.

Juan's celiac story took us from the villus to vitamin K deficiency. Crystal's gallstone will do the same over the weekend. See you Monday for the urinary system — and bring water.

Questions?

See you Monday. The kidneys are calling.

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